SPDTT: Synergizing Prior-Spectrum-Guided Diffusion Model and Target-Aware Transformer for Hyperspectral Target Detection

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***Abstract*—The performance of deep learning models in hyperspectral target detection (HTD) is significantly hampered by two primary challenges: the scarcity of labeled data and the intrinsic spectral variability of targets. To overcome these limitations, this article introduces a novel HTD framework, termed synergizing prior-spectrum-guided diffusion model and target-aware transformer for hyperspectral target detection (SPDTT). First, a prior-spectrum-guided diffusion model (PDM) is designed for data augmentation. By incorporating a customized spectral angle mapping loss, our PDM generates high-fidelity and diverse samples that effectively simulate spectral variations, providing sufficient data for robust model training. Second, a target-aware transformer network is developed for discriminative feature extraction. By utilizing cross-attention, the proposed network deeply embeds prior target knowledge within the feature learning stage. This design guides the model to selectively prioritize the most salient spectral attributes, thereby enhancing its discriminative capability. Furthermore, a composite loss function is introduced to synergistically optimize the network. It enhances interclass separability via a hardness-aware metric, enforces a clear decision boundary through classification supervision, and promotes intraclass compactness to counteract spectral variability. The superiority of the proposed SPDTT method is demonstrated through comprehensive experiments on five publicly available hyperspectral datasets. Results indicate that our approach substantially surpasses both classical and state-of-the-art methods in target detection performance, thereby confirming its effectiveness and robustness.**

***Index Terms*—Cross-attention, diffusion model, hyperspectral target detection (HTD), metric learning, target-aware transformer.**

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I. Introduction

HYPERSPECTRAL imaging synergistically combines 2-D spatial information with spectroscopy. This fusion generates a 3-D hyperspectral image (HSI), where every pixel contains a dense, near-continuous spectrum spanning hundreds of narrow spectral bands, which provides a distinctive fingerprint for identifying the underlying ground materials. This high spectral fidelity enables discerning nuanced material distinctions, supporting applications like classification [1], super-resolution, change detection [2], and target detection [3], [4]. Within these applications, hyperspectral target detection (HTD) stands out as a particularly crucial research area. The principle of HTD is to exploit the unique spectral signatures of specific materials to automatically identify and locate predefined targets within a complex background. This allows for accurate detection even when targets are small, partially occluded, or spectrally similar to the background. Due to their inherent capability for automatic extraction of complex spectral–spatial features, deep learning-based methods have recently emerged as the predominant paradigm in HTD research. Their adoption has consistently yielded enhanced detection accuracy and greater robustness. Consequently, HTD holds significant potential in fields like military reconnaissance and resource exploration, establishing it as a vital research direction. The objective of the HTD task is to develop efficient algorithms capable of suppressing complex backgrounds while exploiting spectral differences to highlight targets. HTD methodologies can be broadly segmented into three principal classes: conventional algorithms, machine learning techniques, and deep learning frameworks.

Early detection methods primarily relied on hypothesis tests or exploited spectral discrepancies between targets and the background. For instance, the matched filter (MF) [5] and the adaptive MF (AMF) [6] both formulate a linear filter to enhance the target's signal response while concurrently mitigating background interference. The adaptive coherence estimator (ACE) [7] and the spectral angle mapper (SAM) detector [8] measure the spectral similarity between each test pixel and the prior target spectrum. Orthogonal subspace projection (OSP) [9] removes background interference and highlights the target with a projection technique. The principle of the constrained energy minimization (CEM) detector [10] is to attenuate background components by reducing the overall output energy. The ensemble-based cascaded CEM (E-CEM) [11] and hierarchical CEM (hCEM) [12] further improve

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robustness and detection accuracy. Although these classical methods laid the foundation for HTD, their reliance on statistical assumptions limits their performance under complex backgrounds and spectral variability.

Machine learning-based methods can be divided into two paradigms: sparse-representation-based and kernel-based. Kernel-based methods enable an effective characterization of nonlinear spectral relationships with projection techniques. Consequently, researchers have extended classical algorithms to their kernel-based versions, such as KCEM [13], KMF [14], and KOSP [15]. These variants consistently achieve superior detection performance over the original linear methods, particularly in scenarios where nonlinearities are dominant. Sparse-representation-based methods [16], [17], [18] reformulate the detection problem as an optimization task under a linear mixing model subject to various regularization constraints. Each pixel can be sparsely represented by a linear combination of background and target atoms.

Exemplifying the successful integration of deep learning into HTD, convolutional neural networks (CNNs) have emerged as a foundational technique. Their prominence is largely attributed to the exceptional ability to automatically learn discriminative features. Du and Li [19] utilized a CNN to determine the class affiliation of each pixel. Zhu et al. [20] proposed two-Stream convolutional networks for hyperspectral target detection (TSCNTD) based on a two-stream design. The proven effectiveness of the transformer architecture has led to its extensive application in HTD. Qin et al. [21] employed a vision transformer (ViT) to capture global characteristics. A combination of a transformer encoder and a momentum encoder was proposed by Wang et al. [22] for robust feature learning. Generative models have also been explored. Zhang et al. [23] proposed HTD-Net, which trains a similarity discrimination network using pixel pairs. Xie et al. [24] developed a detector that incorporates a spectral generative adversarial network (GAN). Li et al. [25] introduced a hierarchical denoising autoencoder (HDAE) for HTD. The background learning based on target suppression constraint for hyperspectral target detection (BLTSC) method [26] isolates the target by subtracting an algorithmically reconstructed background from the original HSI. Luo et al. [27] introduced the SelfMTL method, which integrates self-supervised meta-transfer learning with contrastive representations. Due to the suitability for spectral matching, Siamese network architectures have achieved considerable success in HTD. STTD [28] employs a Siamese network structure built upon the transformer framework. Jiao et al. [29] applied deep metric learning to train a deep Siamese model. Recently, Shen et al. [30] adapted the mamba structure into a Siamese framework.

Despite remarkable progress in deep learning for HTD, several persistent challenges remain.

1) Deep Siamese networks require a large number of balanced sample pairs for training. In practice, however, a primary constraint on model performance stems from the severe class imbalance within training data caused by the scarcity of target samples and the prohibitive expense of hyperspectral data acquisition.
2) Target spectral features often exhibit considerable fluctuations due to varying illumination conditions, atmospheric effects, and sensor noise, resulting in significant spectral variability for the target across different scenes. This variability poses a major challenge for contemporary methods to reliably separate targets from the background. Consequently, there is a pressing need to design specialized model architectures capable of extracting more discriminative features.

Regarding the issue of insufficient target samples, the key lies in effectively utilizing the limited prior spectrum and the HSI itself to construct an adequate set of training samples. Contemporary solutions can be broadly classified into three paradigms: generation methods, synthesis methods, and cross-domain transfer learning. Generation methods, such as those employing AEs [23] or GANs [31], tend to yield homogeneous samples and are unable to effectively model the subtle spectral fluctuations observed in real target data. Conversely, synthesis approaches, which employ the MixUp technique [32] to linearly blend the prior spectrum with other pixels [29], can suffer from instabilities and may generate unrealistic samples that contaminate the training data. Recently, cross-domain transfer learning has emerged as a promising direction, transferring knowledge from data-rich sources. For instance, Jiang et al. [33] established a systematic framework for cross-domain classification utilizing diverse supervision signals, while Feng et al. [34] developed a transformer-based cross-domain few-shot learning (TCFSL) method for HTD that aligns cross-domain features via attention mechanisms. However, despite their effectiveness, these methods fundamentally hinge on the availability of compatible external datasets, making them susceptible to negative transfer caused by spectral shifts. Consequently, exploring the internal potential of the data remains crucial. Diffusion models [35], [36] have recently drawn substantial attention for their successful application to a variety of hyperspectral tasks [37], [38], [39], [40]. The aforementioned studies highlight the potential of diffusion models for producing samples of notable superiority. To address the challenges of sample scarcity and spectral variability, inspired by the latent diffusion model [41], we design a prior-spectrum-guided diffusion model (PDM) for sample generation. In contrast to the cross-domain transfer paradigm, the proposed PDM focuses on exploiting the internal potential of the target scene itself. By utilizing the prior target spectrum as a condition to guide the generation process via a cross-attention mechanism, we ensure that generated samples are relevant to the target with realistic spectral diversity. This strategy effectively circumvents the dependence on external data and the associated risks of domain shift, while providing sufficient high-fidelity data for robust training. Furthermore, to prevent samples from becoming overly similar to the prior and failing to capture realistic variations, we incorporate a SAM loss, which constrains the model to focus on the spectral geometric connection between real samples and the prior spectrum, rather than their absolute values. The synergistic design leads to generated samples that maintain high fidelity while preserving realistic spectral diversity.

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For discriminative feature extraction, a tradeoff exists between CNN-based and transformer-based approaches. While CNNs demonstrate exceptional performance in capturing local information [19], [20], they face limitations in modeling global dependencies. Conversely, although transformers can address this limitation, they tend to neglect local details, thereby reducing feature discriminability [21], [22]. Hybrid approaches have been explored, such as combining groupwise spectral embedding with a transformer encoder [29]. Nevertheless, these methods may still fail to prioritize the most discriminative spectral bands, leading to limited performance when targets and backgrounds are spectrally similar. In response to the aforementioned challenges, a target-aware transformer architecture is designed. It processes the test spectrum and the prior spectrum in parallel, enabling deep information interaction via a cross-attention module [42]. This design allows the network to capture global dependencies while actively focusing on features critical for target discrimination, thus effectively suppressing background interference and enhancing sensitivity to subtle spectral differences.

To achieve a precise separation between targets and the background, we propose a composite target-aware metric (TAM) loss with a hardness-aware weighting scheme to dynamically focus on hard samples. Furthermore, a binary cross-entropy (BCE) loss is employed to sharpen the decision boundary, while a target compactness (TC) term explicitly enforces feature aggregation against spectral variability.

Based on the preceding analysis, this work proposes an HTD method, denoted as synergizing prior-spectrum-guided diffusion model and target-aware transformer for hyperspectral target detection (SPDTT). To address data scarcity and spectral variability, we first design a PDM to generate diverse target samples. Subsequently, we construct a target-aware transformer network that leverages a cross-attention mechanism, enabling the fusion of prior knowledge into the feature learning pipeline. Furthermore, we introduce the TAM loss with a hardness-aware triplet (HAT) mechanism to separate target and background samples while enhancing the intraclass compactness of target features. The primary contributions of this study are summarized as follows.

1) A specially designed PDM is proposed to address critical limitations in conventional sample generation. Conventional conditional models use the condition solely for feature extraction during denoising. However, these approaches often lead to samples with insufficient diversity that fail to simulate realistic spectral variability. In contrast, our PDM employs the prior spectrum in a dual-constraint capacity. In addition to guiding the feature extraction via cross-attention, the prior spectrum is also explicitly incorporated into a novel SAM loss, which constrains the model to learn the intrinsic geometric variations of real samples relative to the prior, rather than merely replicating it. This dual-constraint mechanism enables the generation of high-fidelity, diverse samples that effectively address both sample scarcity and spectral variability.
2) A novel target-aware transformer network is proposed for discriminative feature extraction. By leveraging a cross-attention mechanism, the network continuously incorporates prior information throughout the entire feature extraction process. Unlike conventional methods that compare features postextraction, our network adaptively focuses on the most discriminative spectral features, leading to effective background suppression and enhanced sensitivity to subtle spectral differences.
3) For more precise detection, we introduce a TAM loss. This composite objective function synergistically enhances interclass separability through a hard-sample mining strategy, provides explicit binary supervision to sharpen the decision boundary, and enforces high intraclass consistency for target features. This synergistic design further augments the separability between targets and the background.

The rest of this article details the proposed SPDTT method in Section II, presents an extensive experimental evaluation in Section III, and concludes with a summary and discussion of future work in Section IV.

II. Proposed Method

Throughout this article, we denote scalars as x, vectors as $\mathbf{x}$, matrices as $\mathbf{X}$, and tensors as $\boldsymbol{\mathcal{X}}$. The third-order HSI tensor is signified by $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{h\times w\times l}$, where h, w, and l correspond to the height, width, and the number of spectral bands, respectively. The prior target spectrum is denoted by $\mathbf{d} \in \mathbb{R}^{l\times 1}$. The HSI comprises a total of $n = h \times w$ pixels.

A. Preliminaries

1) Denoising Diffusion Probabilistic Model: The denoising diffusion probabilistic model (DDPM) approximates the true data distribution by learning a denoising process. It is realized through a diffusion stage and a denoising stage.

In the diffusion stage, an initial data sample $\mathbf{x}_0 \in \mathbb{R}^{l\times 1}$ is incrementally corrupted with Gaussian noise over T discrete time steps via a fixed Markov chain. The intensity of the noise at each step is controlled by $\{\beta_t \in (0,1)\}_{t=1}^{\mathrm{T}}$. Leveraging the reparameterization trick allows for the direct computation of the noisy data $\mathbf{x}_t$ from the initial data $\mathbf{x}_0$ at any time step t. With $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$, this diffusion stage is formulated as

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon} \tag{1}$$

where $\boldsymbol{\epsilon}$ denotes the standard Gaussian noise, formulated as $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. This establishes a direct relationship between $\mathbf{x}_0$ and $\mathbf{x}_t$, defined by the $q(\mathbf{x}_t|\mathbf{x}_0)$

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}\left(\mathbf{x}_t;\ \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}\right) \tag{2}$$

where the notation $\mathcal{N}(\cdot;\boldsymbol{\mu},\boldsymbol{\Sigma})$ specifies a Gaussian distribution, and $\boldsymbol{\mu}$, $\boldsymbol{\Sigma}$, and $\mathbf{I}$ represent the mean, covariance, and the identity matrix, respectively. As t increases from 1 to T, the value of $\bar{\alpha}_t$ approaches 0. This implies that the weight of the original signal $\mathbf{x}_0$ gradually diminishes, while the weight of the noise correspondingly increases. When $t = T$, $\mathbf{x}_T$ approximates a pure Gaussian noise.

The denoising stage initiates with pure noise $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and then iteratively denoises it to reconstruct a sample conforming to the true data distribution. This process is likewise

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a Markov chain and is typically parameterized by a U-Net model, denoted as p_θ, and can be expressed as

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\left(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t)\right) \tag{3}$$

where θ denotes the model parameters. This stage aims to approximate the true but intractable posterior distribution. The DDPM article [35] demonstrated that the network only needs to focus on the mean $\boldsymbol{\mu}_\theta(\mathbf{x}_t, t)$, which can be further reparameterized as a function of a noise prediction $\boldsymbol{\epsilon}_\theta$. This relationship is given by

$$\boldsymbol{\mu}_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right) \tag{4}$$

where $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ represents the noise estimate conditioned on the noisy sample $\mathbf{x}_t$ and time step t. Consequently, the loss of DDPM can be formulated as

$$L_{\text{DDPM}} = \mathbb{E}_{t,\mathbf{x}_0,\boldsymbol{\epsilon}}\left[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\|^2\right] \tag{5}$$

where $\mathbb{E}$ denotes the expectation. The inference process commences with a random noise $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. By iteratively executing the denoising steps, this network can produce a new sample that adheres to the learned data distribution.

2) Conditional Diffusion Model: While the standard DDPM excels at generating high-fidelity samples from the learned data distribution, it lacks precise control over the generation process. Consequently, the resulting samples are random and may not align with specific, desired attributes. To enable controllable generation, DDPM can be extended to a conditional diffusion model (CDM). The key idea is to incorporate additional conditional information, denoted as $\mathbf{c}$, into the model to guide the denoising stage. Specifically, when predicting the noise, the CDM additionally relies on the condition $\mathbf{c}$. Consequently, its objective shifts to learning a conditional noise prediction function $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \mathbf{c})$. The loss is specifically formulated to minimize the noise prediction's expected error when provided with the condition

$$L_{\text{cond}} = \mathbb{E}_{t,\mathbf{x}_0,\boldsymbol{\epsilon},\mathbf{c}}\left[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \mathbf{c})\|^2\right]. \tag{6}$$

By minimizing L_{cond}, the CDM learns to generate samples that follow the conditional probability distribution $p(\mathbf{x}_0|\mathbf{c})$. Subsequently, by providing a specific condition $\mathbf{c}$, the trained model can iteratively generate a sample that adheres to the given constraint, starting from random noise.

A critical limitation persists in this conventional paradigm. The condition $\mathbf{c}$ is typically utilized solely as an input to the denoising network, acting only as soft guidance for feature extraction during the reverse process. The conventional training objective lacks an explicit mechanism to ensure that the generated samples maintain fidelity and variety with respect to the given condition. This can lead to outputs that are overly similar to the prior and fail to capture the subtle, realistic variations essential for HTD tasks. To address this, our proposed method redefines the role of the condition for the specific nuances of HTD. We employ the prior target spectrum $\mathbf{d}$ not only as a conditional input for feature guidance via cross-attention, but also as a direct reference within our novel SAM loss. The SAM loss is explicitly incorporated into the training objective, imposing a hard constraint on the spectral geometry of the generated samples. By imposing a constraint to capture the intrinsic spectral geometric connection between real samples and the prior, rather than merely minimizing a reconstruction error, our PDM ensures the generation of high-quality, realistically diverse samples. This process yields superior training data for the subsequent detection task.

B. Overall Architecture

To address sample scarcity and spectral variability, we propose the SPDTT. As depicted in Fig. 1, our SPDTT method is structured around two interconnected modules: a data augmentation component powered by a PDM and a target detection component built upon a target-aware transformer.

The data augmentation stage generates diverse, high-quality target samples from the HSI $\mathcal{X}$ using limited priors. First, a CEM detector is applied to $\mathcal{X}$ to perform coarse detection, yielding an initial detection map. From this map, a pseudo-target sample set $\mathcal{S}_{\text{pt}}$ and a background sample set $\mathcal{S}$ are constructed. Subsequently, the samples in $\mathcal{S}_{\text{pt}}$ are used as the training data for the PDM, into which the prior spectrum $\mathbf{d}$ is injected as conditional information. After the PDM is trained on these samples, a new set of generated samples, $\mathcal{S}_{\text{gt}}$, is produced through the sampling process. These generated samples are highly correlated with the target in terms of spectral features, yet they exhibit subtle, realistic variations.

In the target detection stage, we design a target-aware transformer network to extract discriminative spectral features. During training, the network employs a weight-sharing, three-branch architecture to process three types of inputs in parallel: a target pixel $\mathbf{x}_{\text{gt}}$, the prior spectrum $\mathbf{d}$, and a background pixel $\mathbf{x}$, which serve as the positive, anchor, and negative elements in a triplet, respectively. An internal cross-attention mechanism drives the network, ensuring profound interaction between the features of any given sample and those of the prior spectrum throughout the entire feature extraction process. Optimized with the proposed TAM loss, the network is driven to learn a structured feature space that pulls target samples closer to the anchor while pushing them further from the background. At inference, the trained model is deployed to analyze each test pixel $\mathbf{x}_{\text{test}}$ within the HSI $\mathcal{X}$. The network operates in a dual-branch mode, where $\mathbf{x}_{\text{test}}$ is fed into one branch and the prior spectrum $\mathbf{d}$ is fed into the other. Crucially, the feature extraction for $\mathbf{x}_{\text{test}}$ is guided by the features from the prior branch via the cross-attention mechanism. The confidence score for the pixel being a target is derived from the cosine similarity between the features of the given pixel and the prior spectrum. Applying this procedure to every pixel in the HSI yields the final detection map $\mathbf{D} \in \mathbb{R}^{h\times w}$.

C. Prior-Spectrum-Guided Diffusion Model

The performance of deep learning approaches in HTD is constrained by the scarcity of labeled target samples. To address this challenge, we propose a sample generation method based on a well-designed PDM. The objective of this method is to leverage the HSI itself, along with limited prior information, to generate an extensive and varied set of target samples, which

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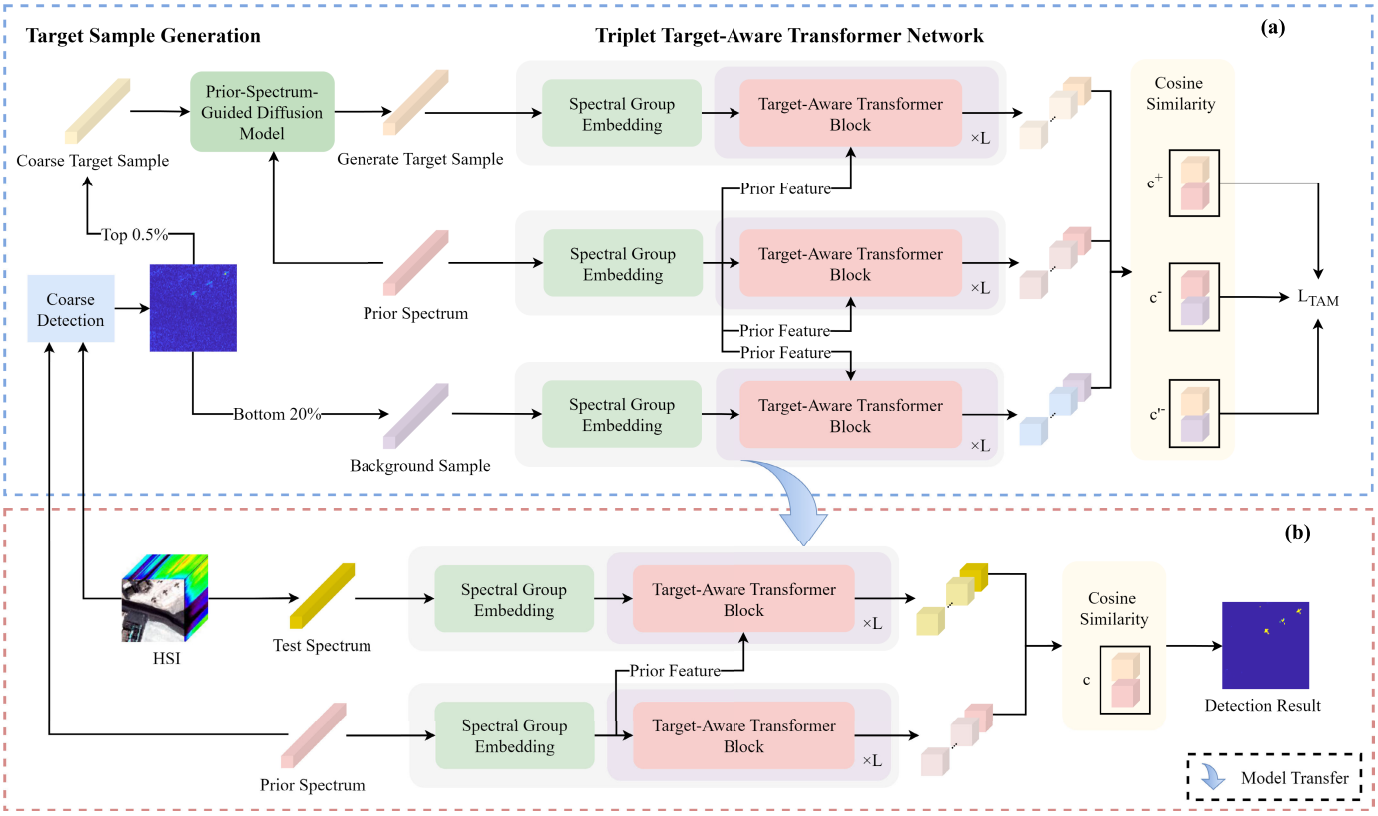


Fig. 1. Architectural diagram of the SPDTT framework. (a) Training stage. (b) Testing stage.

are consistent with the true target distribution while possessing realistic spectral variations.

Notably, our PDM is specifically designed for HTD to detect a predefined target using a single prior spectrum. By utilizing the prior spectrum $\mathbf{d}$ as the guiding condition, the model is trained to generate variations of this specific target. This specialized approach differs from conventional CDMs that learn to generate multiple classes from a large, multiclass dataset. While our method requires retraining for different target classes, the primary strength of our PDM is its capacity to generate superior target data even when initial data are exceptionally scarce, which is a common and practical scenario in hyperspectral applications.

First, the HSI $\mathcal{X}$ and the prior spectrum $\mathbf{d}$ are fed into a CEM detector to perform coarse detection, which generates an initial, less reliable detection map. This map serves as a reference for selecting target and background training samples. Specifically, pixels with the top 0.5% of detection scores are selected as pseudo-target samples, forming the pseudo-target sample set $\mathcal{S}_{\text{pt}} = \{\mathbf{x}_{\text{pt}}^1, \mathbf{x}_{\text{pt}}^2, \ldots, \mathbf{x}_{\text{pt}}^s\}$. Concurrently, pixels with the bottom 20% of scores are chosen as background samples, creating the background sample set $\mathcal{S} = \{\mathbf{x}^1, \mathbf{x}^2, \ldots, \mathbf{x}^m\}$, where $s = n \times 0.5\%$ and $m = n \times 20\%$.

Sample generation is performed by our PDM. A detailed schematic of our PDM is provided in Fig. 2. Initially, we randomly select a spectral vector $\mathbf{x}_0$ from the pseudo-target sample set $\mathcal{S}_{\text{pt}}$. A time step t is randomly assigned to it, and the corresponding noisy sample $\mathbf{x}_t$ is constructed according to (1). The noisy sample is then fed into the denoising network.

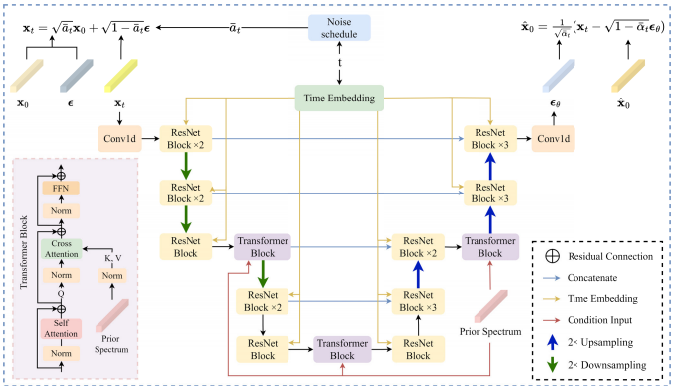


Fig. 2. Illustration of the PDM.

Concurrently, a specialized time embedding module projects the time step t into a high-dimensional feature representation, providing the network with contextual information about the current noise level. The globally shared prior target spectrum $\mathbf{d}$ serves as the condition information, guiding the sample generation direction of the network.

The denoising network adopts a standard U-Net architecture. In the encoder path, the feature sequence passes through multiple stages composed of ResNet blocks and downsampling layers to progressively extract deeper features. We additionally embed transformer blocks in the deeper layers and the bottleneck of the network to model global dependencies among the spectral features. A symmetric decoder path then gradually

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restores the feature resolution via upsampling layers and ResNet blocks, and fuses multiscale information to preserve details by concatenating its features with those from the corresponding encoder levels through skip connections. The conditional guidance mechanism is realized through these transformer blocks. Specifically, the prior target spectrum $\mathbf{d}$ is layer-normalized and then linearly projected within these blocks to generate the key (K) and value (V) matrices. In the subsequent cross-attention step, the K and V matrices interact with the query (Q) matrix, which is computed from the sample's own features. This mechanism ensures that the denoising process can dynamically reference the target's characteristics, thereby achieving precise control over the generation direction. Finally, the network outputs a prediction of the noise $\boldsymbol{\epsilon}_\theta$.

Conventional CDMs typically rely on the mse loss. However, in scenarios involving HSI with distinct physical meanings, relying solely on mse loss presents inherent limitations. First, mse loss disregards the overall geometric structure of the spectrum, which offers more discriminative information than individual bands and is more robust to external factors like illumination changes. Another significant drawback of the mse loss is its tendency to encourage the generation of samples with low diversity, which adhere too closely to the prior spectrum $\mathbf{d}$ and do not reflect real-world spectral variability. Our proposed SAM loss, L_{SAM}, is specifically designed to address this shortcoming. We impose a higher level spectral geometry constraint. Rather than replicating real samples, our approach utilizes the spectral angle to guide it toward learning the inherent patterns of spectral variation. To construct L_{SAM}, we need to recover an estimate of the original sample $\hat{\mathbf{x}}_0$ from the model's output $\boldsymbol{\epsilon}_\theta$. As defined in (2), $q(\mathbf{x}_t|\mathbf{x}_0)$ follows a Gaussian distribution. Therefore, its log-probability gradient is formulated as

$$\nabla_{\mathbf{x}_t} \log q\left(\mathbf{x}_t|\mathbf{x}_0\right) = -\frac{\boldsymbol{\epsilon}}{\sqrt{1-\bar{\alpha}_t}}. \tag{7}$$

In diffusion models, the probability distribution of an observed sample $\mathbf{x}_t$ is referred to as the marginal density, denoted as $q_t(\mathbf{x}_t)$. As demonstrated by Niedoba et al. [43], its log-probability gradient is given by

$$\nabla_{\mathbf{x}_t} \log q_t\left(\mathbf{x}_t\right) = \mathbb{E}_{q(\mathbf{x}_0|\mathbf{x}_t)}\left[\nabla_{\mathbf{x}_t} \log q\left(\mathbf{x}_t|\mathbf{x}_0\right)\right]. \tag{8}$$

We can derive an expression for $\nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ by substituting (7) into (8) and subsequently rearranging the terms

$$\nabla_{\mathbf{x}_t} \log q_t\left(\mathbf{x}_t\right) = -\frac{1}{\sqrt{1-\bar{\alpha}_t}}\mathbb{E}\left[\boldsymbol{\epsilon}|\mathbf{x}_t\right] \tag{9}$$

where $\mathbb{E}[\boldsymbol{\epsilon}|\mathbf{x}_t]$ represents the expected value of $\boldsymbol{\epsilon}$ conditioned on $\mathbf{x}_t$. Chung et al. [44] demonstrated the relationship between $\hat{\mathbf{x}}_0$ and $\nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ as

$$\hat{\mathbf{x}}_0 := \mathbb{E}\left[\mathbf{x}_0|\mathbf{x}_t\right] = \frac{1}{\sqrt{\bar{\alpha}\left(t\right)}}\left(\mathbf{x}_t + \left(1-\bar{\alpha}\left(t\right)\right)\nabla_{\mathbf{x}_t} \log q_t\left(\mathbf{x}_t\right)\right). \tag{10}$$

By substituting (9) into (10), $\hat{\mathbf{x}}_0$ can be expressed in terms of $\mathbb{E}[\boldsymbol{\epsilon}|\mathbf{x}_t]$

$$\hat{\mathbf{x}}_0 = \frac{1}{\sqrt{\bar{\alpha}_t}}\left(\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\mathbb{E}\left[\boldsymbol{\epsilon}|\mathbf{x}_t\right]\right). \tag{11}$$

In practice, by replacing $\mathbb{E}[\boldsymbol{\epsilon}|\mathbf{x}_t]$ with the model's noise prediction $\boldsymbol{\epsilon}_\theta$, we can approximate $\hat{\mathbf{x}}_0$ as

$$\hat{\mathbf{x}}_0 = \frac{1}{\sqrt{\bar{\alpha}_t}}\left(\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}_\theta\right). \tag{12}$$

This equation establishes a direct link between $\hat{\mathbf{x}}_0$ and the model's output $\boldsymbol{\epsilon}_\theta$, facilitating the integration of spectral geometric constraints.

L_{SAM} does not directly minimize the discrepancy between $\hat{\mathbf{x}}_0$ and $\mathbf{x}_0$, but rather the difference between their respective spectral angles with the prior spectrum $\mathbf{d}$. The L_{SAM} is expressed as

$$L_{\text{SAM}} = \mathbb{E}\left[\left(\text{SAM}\left(\hat{\mathbf{x}}_0, \mathbf{d}\right) - \text{SAM}\left(\mathbf{x}_0, \mathbf{d}\right)\right)^2\right] \tag{13}$$

$$\text{SAM}\left(\hat{\mathbf{x}}_0, \mathbf{d}\right) = \arccos\left(\frac{\hat{\mathbf{x}}_0 \cdot \mathbf{d}}{\|\hat{\mathbf{x}}_0\|\,\|\mathbf{d}\|}\right) \tag{14}$$

where $\hat{\mathbf{x}}_0 \cdot \mathbf{d}$ denotes the dot product. This design drives the model to learn the intrinsic variation patterns of true samples relative to the prior, simulating realistic spectral variability and yielding samples with greater diversity. Consequently, this approach mitigates the adverse impact of spectral variability on detection performance.

To balance the tradeoff between reconstruction fidelity and spectral geometric consistency, we introduce a hyperparameter ω to weight the SAM loss. The complete loss for the PDM is formulated as

$$L_{\text{PDM}} = L_{\text{cond}} + \omega \cdot L_{\text{SAM}} \tag{15}$$

where ω controls the relative importance of the spectral angle constraint. A proper ω ensures that the generated samples maintain diverse spectral variations while preserving the intrinsic target identity. By minimizing the composite loss, the trained model can generate a large-scale, high-quality, and reasonably diverse set of target samples, denoted as $\mathcal{S}_{\text{gt}} = \{\mathbf{x}_{\text{gt}}^1, \mathbf{x}_{\text{gt}}^2, \ldots, \mathbf{x}_{\text{gt}}^m\}$, under the guidance of the prior spectrum $\mathbf{d}$. Notably, to maintain a balanced input for the detection network, we also set the quantity of generated target samples to be equal to m. A sample triplet consists of $\mathbf{x}_{\text{gt}}$, $\mathbf{d}$, and $\mathbf{x}$, serving as the positive, the anchor, and the negative elements, respectively.

D. Target-Aware Transformer Network

Transformer architectures excel at capturing global spectral context. However, existing transformer-based HTD methods still treat the test spectrum as an isolated sequence during feature extraction, neglecting to incorporate the prior target knowledge. Consequently, the prior is only exploited in a postextraction metric phase, allowing spectrally similar background or noise signatures to distract the network and yield suboptimal features. Building upon the insights from Peng et al. [45] and Jiao et al. [29], we propose a target-aware transformer network that tightly integrates the prior throughout the entire feature extraction pipeline. Guided by the prior, the network selectively emphasizes salient spectral bands, thereby learning more discriminative representations.

During training, the network uses a weight-sharing, three-branch architecture to process the target sample $\mathbf{x}_{\text{gt}}^i$, the

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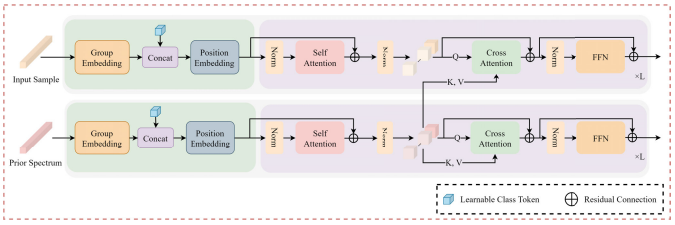


Fig. 3. Schematic of the target-aware transformer network.

prior spectrum $\mathbf{d}$, and the background sample $\mathbf{x}^i$ in parallel. Fig. 3 illustrates the architecture and workflow of the target-aware transformer encoder. Given a hyperspectral sample $\mathbf{x} = \{x^1, x^2, \ldots, x^l\}^{\mathrm{T}} \in \mathbb{R}^{l\times 1}$, we first apply symmetric padding to its spectral dimension to ensure that edge information is processed equally during subsequent sliding window operations. Specifically, we pad $\lfloor w/2 \rfloor$ bands to each end of the spectrum, where w represents the width of the sliding window. The dimension of the padded spectral vector, thus, becomes $l + w$. Subsequently, a groupwise spectral embedding strategy [46] is employed. A sliding window of size w and stride 1 is applied to the padded spectrum, which, due to the symmetric padding, generates a sequence of l spectral groups. Each group is a w-dimensional vector representing the local spectral information of its central band and its neighborhood. Next, a linear projection layer takes these spectral groups and embeds them within a D-dimensional feature space, yielding an initial token sequence $\mathbf{G} = \{\mathbf{g}_1, \mathbf{g}_2, \ldots, \mathbf{g}_l\}^{\mathrm{T}} \in \mathbb{R}^{l\times D}$. For sequence-level feature aggregation, we prepend a learnable class token of dimension $1 \times D$ to the beginning of $\mathbf{G}$. Furthermore, to preserve positional information, a positional encoding is added to the token embeddings. This process results in the final input sequence for the transformer encoder, $\mathbf{Z}_0 \in \mathbb{R}^{(l+1)\times D}$.

The transformer network is composed of L stacked, structurally identical target-aware transformer blocks. The depth of the transformer network L is a critical hyperparameter that directly influences the model's capacity to learn high-level semantic features. Theoretically, a deeper architecture allows for more iterations of cross-attention interaction, which facilitates the extraction of refined features and the modeling of complex, long-range spectral dependencies. This is crucial for distinguishing targets from complex backgrounds. However, in HTD tasks where labeled target samples are limited, excessively increasing L introduces significant risks. Excessive depth not only leads to higher computational costs but also expands the parameter space, making the model more susceptible to overfitting and optimization instability. Therefore, the choice of L represents a strategic tradeoff between feature extraction capability and generalization robustness. Consequently, we determine the optimal value of L through a comprehensive analysis balancing detection accuracy and computational cost, as detailed in Section III-E. Within the ith block, the input feature sequence $\mathbf{Z}_{i-1} \in \mathbb{R}^{(l+1)\times D}$ is first processed by a standard self-attention module. This module generates an intermediate feature sequence $\mathbf{Z}'_i \in \mathbb{R}^{(l+1)\times D}$, which has preliminarily integrated contextual information. This process can be expressed as

$$\mathbf{Z}'_i = \mathrm{SelfAttention}(\mathrm{Norm}(\mathbf{Z}_{i-1})) + \mathbf{Z}_{i-1}. \tag{16}$$

This sequence is then fed into a cross-attention module to obtain enhanced features $\mathbf{Z}''_i \in \mathbb{R}^{(l+1)\times D}$

$$\mathbf{Z}''_i = \mathrm{CrossAttention}\left(\mathrm{Norm}\left(\mathbf{Z}'_i\right), \mathrm{Norm}\left(\mathbf{Z}_i^{p'}\right)\right) + \mathbf{Z}'_i \tag{17}$$

where $\mathbf{Z}_i^{p'}$ denotes the corresponding feature sequence in the prior branch. The generation process of $\mathbf{Z}_i^{p'}$ is identical to that of $\mathbf{Z}'_i$. Through the information interaction, the feature representation of the current branch is dynamically adjusted under the guidance of the prior spectrum. This adaptively enhances target-related discriminative features while suppressing irrelevant background interference.

Finally, the feature sequence enhanced by cross-attention is passed through a feed-forward network (FFN) for nonlinear transformation. This transformation boosts the model's expressive capability and yields the final block output, $\mathbf{Z}_i \in \mathbb{R}^{(l+1)\times D}$

$$\mathbf{Z}_i = \mathrm{FFN}\left(\mathrm{Norm}\left(\mathbf{Z}''_i\right)\right) + \mathbf{Z}''_i. \tag{18}$$

After being processed by L blocks, we obtain the final feature sequence $\mathbf{Z}_L = \{\mathbf{z}_L^{\mathrm{cls}}, \mathbf{z}_L^1, \ldots, \mathbf{z}_L^l\} \in \mathbb{R}^{(l+1)\times D}$. The output vector corresponding to the class token, $\mathbf{z}_L^{\mathrm{cls}}$, is then singled out from the final sequence. This vector acts as the global representation for the input hyperspectral sample.

E. TAM Loss Function

To guide the proposed target-aware transformer network to learn a feature space that can both clearly separate targets from the background and ensure high intraclass compactness for target samples, we design a novel composite loss function, termed the TAM loss. The TAM loss provides comprehensive guidance for network optimization by integrating metric learning, classification supervision, and intraclass constraints. A detailed explanation is provided in the following.

1) HAT Loss: Triplet loss is a classic method in metric learning. However, the conventional triplet loss treats all samples equally, wasting computational resources on simple samples. Our proposed HAT loss is specifically designed to mitigate this inefficiency. It can dynamically assign a weight to each sample triplet based on its discrimination difficulty, enabling the model to focus on samples that are hard to distinguish. For the ith sample triplet, the features extracted by the network for the target sample $\mathbf{x}_{\mathrm{gt}}^i$, the prior spectrum $\mathbf{d}$, and the background sample $\mathbf{x}^i$ are denoted as $\mathbf{f}(\mathbf{p}_i)$, $\mathbf{f}(\mathbf{a})$, and $\mathbf{f}(\mathbf{n}_i)$, respectively. Our loss function is formulated based on the cosine similarity, whose definition is given by

$$c(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\,\|\mathbf{v}\|} \tag{19}$$

where $\mathbf{u}$ and $\mathbf{v}$ represent the two vectors of identical dimensionality, and $c \in [-1, 1]$. We pair the features within each sample triplet to obtain the following three similarity scores:

$$\begin{aligned} c_i^+ &= |c(\mathbf{f}(\mathbf{p}_i), \mathbf{f}(\mathbf{a}))| \\ c_i^- &= |c(\mathbf{f}(\mathbf{n}_i), \mathbf{f}(\mathbf{a}))| \\ c_i'^- &= |c(\mathbf{f}(\mathbf{p}_i), \mathbf{f}(\mathbf{n}_i))|. \end{aligned} \tag{20}$$

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The objective is to enforce a margin between the positive similarity and the larger of the two negative similarities. We select the maximum of the two negative scores as the triplet's overall negative similarity, denoted as $c_i^* = \max(c_i^-, c_i'^-)$. Then, the triplet loss term is given by

$$l_i = \max\left(\mu + c_i^* - c_i^+, 0\right) \tag{21}$$

where μ is a predefined margin value. Subsequently, we define the hardness of this triplet as $h_i = 1 - c_i^+$ and utilize a softmax function to convert the hardness scores of all samples in the batch into normalized weights $\mathbf{w} = \{w_1, w_2, \ldots, w_b\}^{\mathrm{T}} \in \mathbb{R}^{b\times 1}$. The HAT loss is computed by aggregating the base losses of all triplets, where each base loss is weighted by its hardness score. This can be formulated as

$$L_{\mathrm{HAT}} = \sum_{i=1}^{b} w_i \cdot l_i \tag{22}$$

where b denotes the number of triplets. Minimizing HAT loss encourages the positive samples to be as close as possible to the prior spectrum, while pushing the negative samples far from the positive samples and the prior spectrum. Furthermore, the dynamic weighting scheme actively steers the model's attention toward hard samples.

2) BCE Loss: A limitation of the HAT loss is that it operates solely on the relative distances among samples, without leveraging their ground-truth class labels. We rectify this omission by incorporating a BCE loss. This component directly transforms the similarity metric task into a binary classification framework. Specifically, we assign binary labels to the similarity scores, where $y_i^+ = 1$ for the positive pair and $y_i^- = 0$ for the negative pair. The BCE loss is then formulated as

$$L_{\mathrm{BCE}} = -\frac{1}{2b}\sum_{i=1}^{b}\left\{y_i^+ \cdot \log\left[\sigma\left(c_i^+\right)\right] + \left(1 - y_i^-\right)\cdot \log\left[\sigma\left(1 - c_i^-\right)\right]\right\} \tag{23}$$

where $\sigma(\cdot)$ denotes the sigmoid function.

3) TC Loss: Although the HAT loss and BCE loss work together to effectively separate target and background samples, they primarily focus on interclass separability. However, the feature representations of target samples can be scattered, which blurs the decision boundary and degrades detection stability. The TC loss is proposed specifically to counteract this effect. It pulls the features of all target samples closer to the prior target feature, as defined in the following formula:

$$L_{\mathrm{TC}} = \frac{1}{b}\sum_{i=1}^{b}\left(1 - c_i^+\right)^2. \tag{24}$$

In summary, we combine the three aforementioned loss terms to form our TAM loss, which can be expressed as

$$L_{\mathrm{TAM}} = L_{\mathrm{HAT}} + \alpha_1 \cdot L_{\mathrm{BCE}} + \alpha_2 \cdot L_{\mathrm{TC}} \tag{25}$$

where the balancing hyperparameters α_1 and α_2 regulate the feature space. Specifically, α_1 controls the BCE loss intensity, which requires a large value to enforce a sharp decision boundary for background suppression. In contrast, α_2 functions as a regularization constraint for intraclass aggregation. A smaller value of α_2 is preferred to maintain compactness while preventing feature collapse and preserving spectral diversity.

TABLE I
SUMMARY OF KEY HYPERPARAMETERS USED IN THE EXPERIMENTS

Parameter Description	Symbol	Value
Learning Rate	lr	1×10^{-4}
Number of Time Steps	T	1000
Number of Transformer Blocks	L	4
HAT Loss Margin	μ	1
SAM Loss Weight	ω	1
BCE Loss Weight	α_1	10
TC Loss Weight	α_2	0.01

III. EXPERIMENTAL EVALUATIONS

A. Datasets

The proposed method is evaluated on five public hyperspectral datasets. Their details and included targets are summarized as follows.

1) *San Diego [47]:* This dataset covers a 100×100 pixel area and features 224 spectral bands encompassing the wavelength range from 370 to 2510 nm. The targets are three aircraft. After removing noisy bands, we utilized 186 effective bands.
2) *Airport [48]:* This dataset was acquired by the AVIRIS sensor. It features 100×100 pixels and a ground sampling distance of 3.4 m. After removing noisy bands, 191 spectral bands are used, with three aircraft as the targets.
3) *Urban-I [48]:* The AVIRIS sensor collected this dataset, which captures a 100×100 pixel area and features 17.2-m resolution. After removing problematic bands, 207 bands are used, including 20 oil tanks of various sizes serving as targets.
4) *Urban-II [48]:* This dataset provides a 100×100 pixel area and 7.1-m ground resolution. It contains 205 spectral bands, each characterized by a 10-nm resolution. The scene features 25 differently sized oil tanks as the targets.
5) *Indian Pines [49]:* This dataset provides a 145×145 pixel scene and a 20-m ground resolution. After the elimination of 20 corrupted bands, the image consists of 200 spectral bands. For this study, we selected class 16, which contains fewer pixels, as the target, while all other pixels were treated as background.

B. Experimental Setup

We evaluate SPDTT against several widely recognized HTD methods. These include four classical methods: CEM [10], ACE [7], CSCR [17], and hCEM [12]. For deep learning approaches, the comparison includes BLTSC [26], HTD-IRN [50], TSTTD [29], and GODSNCR [51]. All source code was downloaded from the authors' public repositories, and the hyperparameters were tuned to the optimal settings as guided

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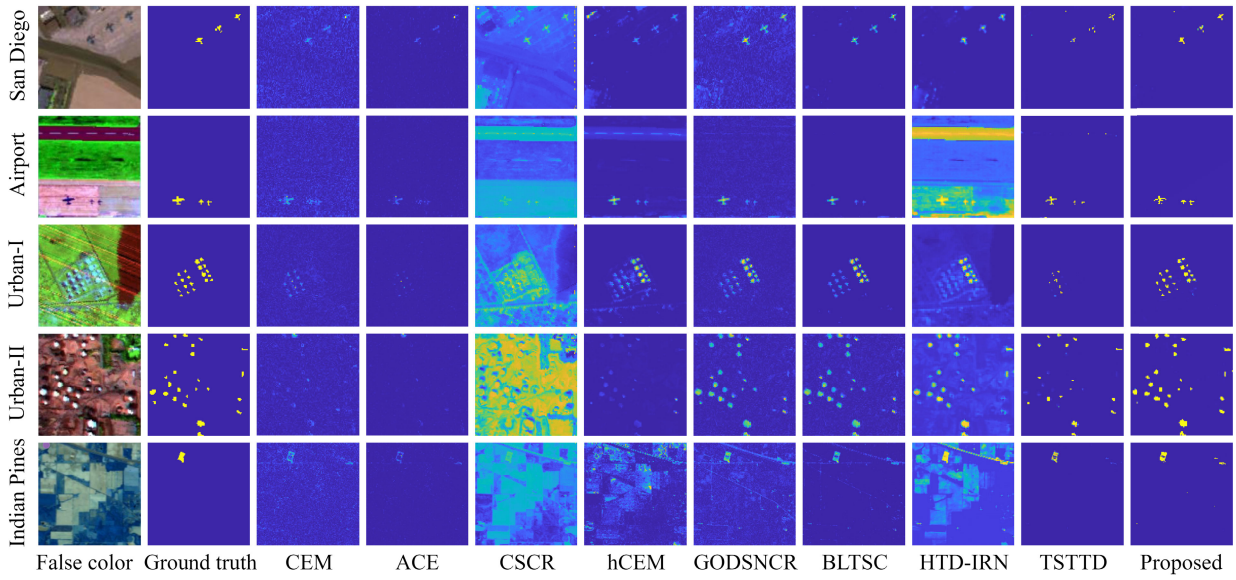


Fig. 4. Visualization of detection results for all evaluated approaches.

by their respective articles. For fair comparison, all approaches utilized the same number of training samples and the same prior spectrum $\mathbf{d}$. Specifically, to ensure the representativeness of the prior information and mitigate the impact of outliers, $\mathbf{d}$ was selected as the ground-truth target pixel that minimizes the Euclidean distance to the average target spectrum of the entire scene. Let $\mathcal{R} = \{\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_{N_{\text{gt}}}\}$ denote the set of all N_{gt} ground-truth target pixels available in the scene, and let $\bar{\mathbf{r}} = 1/N_{\text{gt}} \sum_{i=1}^{N_{\text{gt}}} \mathbf{r}_i$ represent their global mean spectral vector. The selection criterion for $\mathbf{d}$ is mathematically defined as

$$\mathbf{d} = \arg\min_{\mathbf{r}_i \in \mathcal{R}} \|\mathbf{r}_i - \bar{\mathbf{r}}\|. \tag{26}$$

Unlike using the synthesized mean spectrum $\bar{\mathbf{r}}$ directly, this strategy ensures that $\mathbf{d}$ remains a physically real instance, preserving realistic spectral characteristics while serving as the robust statistical centroid of the target class. Regarding the baseline of generative models, all datasets were normalized to the range of $[0, 1]$ to ensure a consistent data scale. For the diffusion-based models, DDPM and PDM, an identical number of time steps $T = 1000$ was employed to align the denoising process. All experiments used Python 3.12.3 and PyTorch 2.4.0 on an AMD Ryzen 9 7945HX CPU and NVIDIA GeForce RTX 4060 GPU. For the implementation of SPDTT, we set the margin $\mu = 1$. The network was optimized using the Adam optimizer with a learning rate of 1×10^{-4}. Furthermore, to determine the optimal model configuration, we conducted a comprehensive sensitivity analysis on key hyperparameters in Section III-E. This analysis specifically investigated the number of transformer blocks L, the PDM loss weight ω, and the balancing coefficients α_1 and α_2 in TAM loss. To provide a clear overview of the experimental configuration, we summarize the hyperparameters in Table I.

C. Evaluation Criteria

The performance is assessed through visual detection maps and receiver operating characteristic (ROC) curves [52], which are generated by varying a threshold τ. A true positive (TP) occurs when a target pixel's detection value exceeds τ, whereas a false negative (FN) is a target pixel whose value falls below τ. Conversely, a false positive (FP) is a background pixel incorrectly classified as a target, and a true negative (TN) is a background sample correctly identified as such. To assess the performance of different methods from multiple perspectives, we utilize different 2-D ROC curves. Target detection integrity is evaluated using the (τ, P_d) curve, which is expected to be as close to the top right. Background suppression capability is measured by the (τ, P_f) curve, which should be as close to the bottom left, and the (P_f, P_d) curve summarizes the overall performance. It is desired to be as close as the top left. Detection rate P_f and false alarm rate P_d are determined at each threshold via the following formulas:

$$P_d = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad P_f = \frac{\text{FP}}{\text{FP} + \text{TN}}. \tag{27}$$

A quantitative assessment is performed by calculating the area under these 2-D ROC curves, resulting in three area under curve (AUC) scores. Specifically, a higher $\text{AUC}(P_f, P_d)$ reflects better comprehensive performance. Similarly, a higher $\text{AUC}(\tau, P_d)$ indicates more robust detection integrity. The ideal score for both is 1. Conversely, a lower $\text{AUC}(\tau, P_f)$ represents better background suppression capability, with an optimal value of 0. Furthermore, to more comprehensively evaluate the method's performance, we integrate these three AUC values into two composite metrics, denoted as AUC_{OA} and AUC_{SNPR}. Their formulas are given by

$$\text{AUC}_{\text{OA}} = \text{AUC}(P_f, P_d) + \text{AUC}(\tau, P_d) - \text{AUC}(\tau, P_f) \tag{28}$$

$$\text{AUC}_{\text{SNPR}} = \frac{\text{AUC}(\tau, P_d)}{\text{AUC}(\tau, P_f)}. \tag{29}$$

For these two metrics, higher values indicate superior detection performance.

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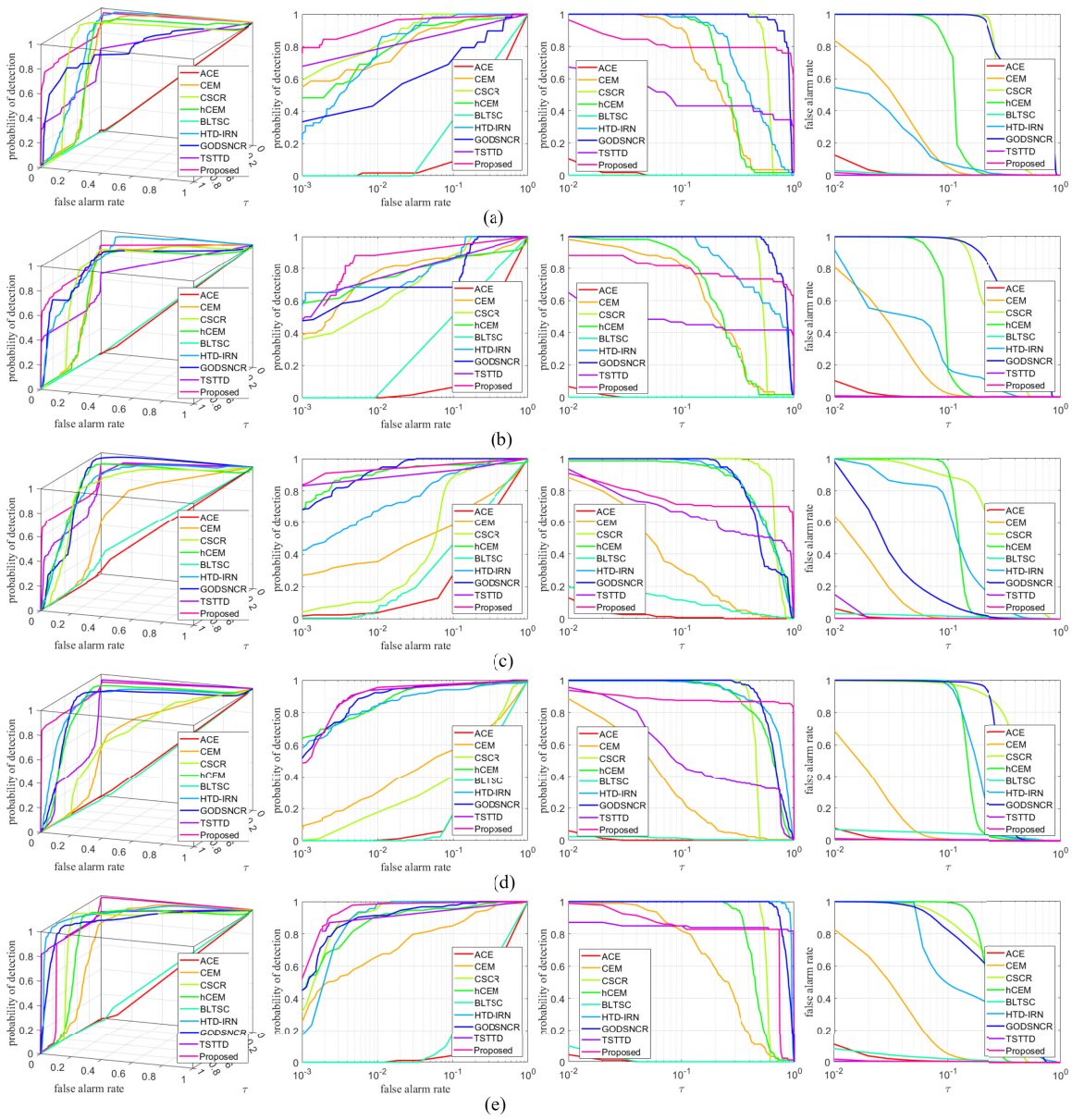


Fig. 5. ROC curves for various comparative methods on five datasets. From left to right are the 3-D ROC curve and 2-D ROC of (P_f, P_d), (τ, P_d), and (τ, P_f) curves, respectively. (a) San Diego. (b) Airport. (c) Urban-I. (d) Urban-II. (e) Indian pines.

D. Comparative Results and Analysis

Fig. 4 qualitatively compares the detection performance for all evaluated methods across the five datasets. Classical detectors, such as CEM and ACE, can locate targets in some simple scenarios, but their detection maps suffer from high levels of residual background noise and attenuated target responses, making it difficult to form clear target contours. ACE, in particular, fails to effectively suppress the background in most scenes, leading to an overall high response across the detection map. As an advancement over CEM, hCEM exhibits stronger background suppression in most scenarios. However, its target detection integrity remains insufficient, often resulting in the loss of target edge details or the omission of some target parts. In contrast, deep learning-based methods generally excel at highlighting targets. For instance, HTD-IRN and GODSNCR generally demonstrate better target detection integrity. However, they struggle with effective background suppression in the face of complex backgrounds, resulting in numerous false alarms. BLTSC and TSTTD successfully address the issue of excessive background response, achieving smoother background suppression. Nevertheless, their target response intensity and detection completeness are still lacking, causing them to misclassify targets as background. By comparison, our proposed method demonstrates the best overall detection performance, with its results exhibiting both strong background suppression and excellent target detection integrity. Two synergistic designs are central to our approach, culminating in enhanced performance. First, the PDM generates a sufficient volume of high-quality training samples, enhancing the model's robustness to spectral variability. Second, our proposed target-aware transformer architecture fully

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TABLE II
QUANTITATIVE PERFORMANCE RESULTS OBTAINED BY ALL METHODS ON FIVE DATASETS. OPTIMAL AND SUBOPTIMAL RESULTS ARE INDICATED BY BOLD AND UNDERLINED TEXT, RESPECTIVELY

Dataset		CEM	ACE	CSCR	hCEM	GODSNCR	BLTSC	HTD-IRN	TSTTD	Proposed
San Diego	AUC(P_f, P_d)	0.9785	0.9794	0.9942	0.9663	0.9043	0.9647	0.9910	0.9146	**0.9970**
	AUC(τ, P_d)	0.2545	0.1085	0.5963	0.2786	**0.8890**	0.0917	0.4511	0.4128	0.7851
	AUC(τ, P_f)	0.0395	0.0043	0.3045	0.1146	0.5723	0.0010	0.0358	**0.0009**	0.0018
	AUC$_{OA}$	1.1935	1.0836	1.2861	1.1303	1.2210	1.0553	1.4063	1.3265	**1.7803**
	AUC$_{SNPR}$	6.4430	25.2326	1.9586	2.4311	1.5534	89.3689	12.5941	**458.6667**	436.1667
	Time	0.0400	**0.0084**	3.0370	0.1823	18.1158	0.1124	1.2019	2.3113	2.6000
Airport	AUC(P_f, P_d)	0.9309	0.9079	0.9695	0.9098	0.9521	0.9648	0.9739	0.9922	**0.9993**
	AUC(τ, P_d)	0.2446	0.1122	0.5375	0.2553	**0.8256**	0.2183	0.6335	0.4349	0.7371
	AUC(τ, P_f)	0.0338	0.0039	0.3139	0.0932	0.4028	0.0014	0.0739	**0.0009**	0.0038
	AUC$_{OA}$	1.1417	1.0162	1.1930	1.0719	1.3749	1.1816	1.5335	1.4262	**1.7326**
	AUC$_{SNPR}$	7.2367	28.7692	1.7120	2.7393	2.0497	151.6803	8.5750	**483.2222**	193.9737
	Time	0.0400	**0.0080**	3.1179	0.1885	18.9228	0.1337	1.3442	2.5510	2.6410
Urban-I	AUC(P_f, P_d)	0.8071	0.6291	0.9369	0.9666	0.9976	0.9710	0.9652	0.9819	**0.9992**
	AUC(τ, P_d)	0.1034	0.0175	0.6882	0.6082	0.5622	0.4021	0.6069	0.5199	**0.8042**
	AUC(τ, P_f)	0.0187	0.0029	0.3193	0.1267	0.0435	0.0031	0.1430	0.0031	0.0010
	AUC$_{OA}$	0.8918	0.6437	1.3058	1.4481	1.5163	1.3701	1.4291	1.4987	**1.8024**
	AUC$_{SNPR}$	5.5294	6.0345	2.1552	4.8003	12.9241	131.3435	4.2455	167.7097	**804.2000**
	Time	0.0445	**0.0147**	6.1162	0.1617	18.8785	0.1216	1.4654	2.9709	2.9450
Urban-II	AUC(P_f, P_d)	0.7793	0.6348	0.7798	0.9790	0.9880	0.9818	0.9692	0.9912	**0.9962**
	AUC(τ, P_d)	0.0847	0.0221	0.4545	0.6155	0.6730	0.5254	**0.7344**	0.3168	0.7231
	AUC(τ, P_f)	0.0212	0.0031	0.3796	0.1555	0.2877	0.0077	0.1856	**0.0014**	0.0021
	AUC$_{OA}$	0.8428	0.6537	0.8547	1.4390	1.3733	1.4995	1.5179	1.3066	**1.7172**
	AUC$_{SNPR}$	3.9953	7.1290	1.1972	3.9582	2.3392	68.3132	3.9563	226.2857	**344.3333**
	Time	0.0396	**0.0096**	6.7153	0.1844	21.2364	0.1152	1.6342	2.8694	2.9440
Indian Pines	AUC(P_f, P_d)	0.9533	0.9046	0.9977	0.9868	0.9932	0.9847	0.9975	0.9981	**0.9983**
	AUC(τ, P_d)	0.2668	0.0906	0.5734	0.4673	0.8783	0.3316	**0.9495**	0.6324	0.8384
	AUC(τ, P_f)	0.0389	0.0043	0.2967	0.2244	0.3411	0.0050	0.2144	0.0041	**0.0022**
	AUC$_{OA}$	1.1812	0.9908	1.2743	1.2297	1.5304	1.3114	1.7326	1.6264	**1.8345**
	AUC$_{SNPR}$	6.8586	21.0698	1.9324	2.0824	2.5749	66.6115	4.4286	154.2439	**381.0909**
	Time	0.0599	**0.0239**	4.7719	0.3269	26.9793	0.2663	2.4264	3.5789	3.8740

leverages the prior target information to isolate the most salient spectral characteristics.

Fig. 5 displays the ROC curves of all approaches, providing a more comprehensive assessment of their detection performance across different scenes. The ROC curves clearly demonstrate the ability of the SPDTT method to achieve complete target detection while maintaining a low false alarm rate. The (P_f, P_d) curves reveal that our method outperforms others in overall detection performance. Although the detection rate of HTD-IRN on the Airport dataset and that of GODSNCR on the Urban-II dataset are slightly higher when P_f is around 10^{-3}, both methods achieve this at the cost of sacrificing background suppression. This tradeoff is also evident from the (τ, P_f) curves, which show that our method is superior in background suppression compared to these two counterparts. The results affirm that our SPDTT method can more effectively distinguish target signatures from background. In conclusion, while our method may not be the absolute top performer in terms of target detection integrity alone, it still surpasses most of the compared methods and, more importantly, achieves a better balance between detection integrity and background suppression. Consequently, our approach consistently secures the best overall detection performance.

The AUC values of all compared methods are summarized in Table II. As shown in the table, the SPDTT consistently attains the top AUC(P_f, P_d) values. This result indicates that the SPDTT excels at maintaining a high detection rate while simultaneously keeping the false alarm rate low. The AUC(τ, P_f) metric reflects a method's background suppression capability, while AUC(τ, P_d) reflects its target detection integrity. As shown in the table, although GODSNCR and HTD-IRN obtain higher AUC(τ, P_d) values on some datasets, their corresponding AUC(τ, P_f) values are also relatively high. This aligns with our qualitative observation of more false alarms in their detection maps, suggesting that they sacrifice a degree of background suppression. Conversely, while TSTTD achieves the best background suppression, its detection integrity remains insufficient. In contrast, our method maintains a high target detection rate while consistently demonstrating strong competitiveness in the AUC(τ, P_f) metric, achieving the optimal or suboptimal values on multiple datasets, particularly on the Urban-I and Indian Pines datasets.

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TABLE III

AUC SCORES FOR DIFFERENT DATA AUGMENTATION METHODS USED ON FIVE DATASETS. OPTIMAL AND SUBOPTIMAL RESULTS ARE INDICATED BY BOLD AND UNDERLINED TEXT, RESPECTIVELY

Dataset		No DA	MixUp	VAE	DDPM	PDM
San Diego	$AUC(P_f, P_d)$	0.8850	0.9146	0.9879	0.9941	**0.9970**
	$AUC(\tau, P_d)$	0.5467	0.4219	0.3637	0.6177	**0.7851**
	$AUC(\tau, P_f)$	0.1403	**0.0009**	0.0163	0.0011	0.0018
Airport	$AUC(P_f, P_d)$	0.6677	0.9522	0.9656	0.9958	**0.9993**
	$AUC(\tau, P_d)$	0.5779	0.4493	0.3623	0.6599	**0.7371**
	$AUC(\tau, P_f)$	0.4031	**0.0009**	0.0183	0.0061	0.0038
Urban-I	$AUC(P_f, P_d)$	0.6455	0.9719	0.9786	0.9947	**0.9992**
	$AUC(\tau, P_d)$	0.3102	0.5147	0.4313	**0.8519**	0.8042
	$AUC(\tau, P_f)$	0.2105	0.0031	0.0349	0.0034	**0.0010**
Urban-II	$AUC(P_f, P_d)$	0.6960	0.9823	0.9729	0.9928	**0.9962**
	$AUC(\tau, P_d)$	0.4930	0.3786	0.6016	**0.7513**	0.7231
	$AUC(\tau, P_f)$	0.0222	**0.0017**	0.0567	0.0048	0.0021
Indian Pines	$AUC(P_f, P_d)$	0.6480	0.8868	0.9831	0.9957	**0.9983**
	$AUC(\tau, P_d)$	0.6035	0.6342	0.6181	0.7911	**0.8384**
	$AUC(\tau, P_f)$	0.2204	0.0072	0.0233	0.0086	**0.0022**

This result once again confirms the excellent background suppression capability of SPDTT. The composite evaluation metrics AUC_{OA} and AUC_{SNPR} further corroborate the superiority of our method. SPDTT obtains the highest AUC_{OA} values on all datasets, and its AUC_{SNPR} results are also either optimal or suboptimal, showcasing its powerful and comprehensive detection performance in complex environments. In terms of computational efficiency, Table II also records the inference time of each method on the various datasets. It is evident that the classical methods achieve the fastest detection speeds, due to their concise mathematical models. In contrast, deep learning-based methods typically require longer inference times due to their complex network architectures and extensive parameter computations. However, it is noteworthy that among all the deep learning methods, the inference time of our proposed approach is moderate. It is significantly faster than GODSNCR, for instance, which indicates that our model architecture maintains a degree of computational efficiency while ensuring high performance. In summary, the quantitative analysis results reaffirm the effectiveness of the SPDTT approach. Although some methods may exhibit partial advantages on individual metrics, our approach achieves optimal results on the key comprehensive performance indicators and demonstrates significant superiority in background suppression. This collectively demonstrates its effectiveness and robustness as an advanced HTD method.

E. Ablation Study and Parameter Analysis

1) Effectiveness Evaluation of PDM: To validate the effectiveness of PDM, we compare the impact of our PDM on detection results against a baseline without data augmentation (denoted as No DA), as well as several other spectral data augmentation methods, including MixUp [32], variational autoencoder (VAE) [53], and DDPM [35]. Table III presents the AUC values for these methods across the five datasets. The results show that the PDM consistently outperforms the other data augmentation techniques. Compared to the No DA baseline, all data augmentation techniques significantly improve detection performance, demonstrating the necessity of data augmentation under sample-scarce conditions. The PDM achieves the optimal $AUC(P_f, P_d)$ values on all datasets and also shows superior performance in $AUC(\tau, P_f)$ and $AUC(\tau, P_d)$, which strongly demonstrates that by introducing the prior target spectrum as a conditional guide and incorporating the proposed SAM loss, L_{SAM}, as a constraint, our PDM can generate high-quality target samples with reasonable diversity. Consequently, the finally trained detector achieves the best overall performance.

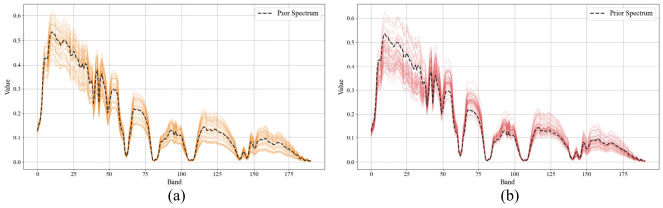


Fig. 6. Visual comparison of spectral curves on the Urban-I dataset. The black dashed line indicates the prior spectrum. (a) Real target spectra. (b) Generated target spectra.

TABLE IV

AUC SCORES FOR PDM WITH AND WITHOUT L_{SAM} ON FIVE DATASETS. OPTIMAL RESULTS ARE INDICATED BY BOLD TEXT

Dataset		L_{SAM} ×	L_{SAM} √
San Diego	$AUC(P_f, P_d)$	0.9957	**0.9970**
	$AUC(\tau, P_d)$	**0.8065**	0.7851
	$AUC(\tau, P_f)$	0.0032	**0.0018**
Airport	$AUC(P_f, P_d)$	0.9980	**0.9993**
	$AUC(\tau, P_d)$	0.6143	**0.7371**
	$AUC(\tau, P_f)$	0.0096	**0.0038**
Urban-I	$AUC(P_f, P_d)$	**0.9996**	0.9992
	$AUC(\tau, P_d)$	0.7845	**0.8042**
	$AUC(\tau, P_f)$	0.0046	**0.0010**
Urban-II	$AUC(P_f, P_d)$	0.9944	**0.9962**
	$AUC(\tau, P_d)$	0.6141	**0.7231**
	$AUC(\tau, P_f)$	**0.0018**	0.0021
Indian Pines	$AUC(P_f, P_d)$	0.9951	**0.9983**
	$AUC(\tau, P_d)$	0.6772	**0.8384**
	$AUC(\tau, P_f)$	0.0081	**0.0022**

2) Effectiveness Evaluation of Spectral Angle Mapping Loss: We conducted an ablation study to verify the effectiveness of the spectral angle mapping loss, L_{SAM}, where we compared the performance of the complete PDM incorporating L_{SAM} against a variant without it. The results are presented in Table IV. Results show that integrating SAM loss yields a significant performance boost. On $AUC(P_f, P_d)$, the complete model outperforms the PDM variant without L_{SAM} on most datasets. The comparison of $AUC(\tau, P_d)$ and $AUC(\tau, P_f)$ also reveals that the complete model's target detection integrity and background suppression capability are superior on most datasets. This suggests that a sole reliance on mse loss tends to generate samples that are highly similar to the prior spectrum but lack diversity, making the model struggle with spectral variability. In contrast, after incorporating the

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TABLE V

QUANTITATIVE PERFORMANCE RESULTS OF DIFFERENT BACKBONES ON FIVE DATASETS. OPTIMAL AND SUBOPTIMAL RESULTS ARE INDICATED BY BOLD AND UNDERLINED TEXT, RESPECTIVELY

Dataset		MLP	ViT	Mamba	U-Net	Proposed
San Diego	Params/M	1.124	1.125	1.124	1.125	1.122
	GFLOPs	0.572	**0.186**	0.255	0.198	0.194
	AUC(P_f, P_d)	0.9956	0.9963	0.9965	0.9943	**0.9970**
	Time	**1.589**	2.368	3.546	2.294	2.600
Airport	Params/M	1.134	1.139	1.135	1.135	1.133
	GFLOPs	0.774	**0.178**	0.262	0.193	0.196
	AUC(P_f, P_d)	0.9968	0.9988	0.9981	0.9973	**0.9993**
	Time	**1.618**	2.592	3.326	2.165	2.641
Urban-I	Params/M	1.122	1.125	1.122	1.119	1.122
	GFLOPs	0.584	**0.198**	0.286	0.235	0.214
	AUC(P_f, P_d)	0.9962	0.9957	0.9963	0.9977	**0.9992**
	Time	**1.713**	2.798	3.358	2.347	2.945
Urban-II	Params/M	1.144	1.145	1.146	1.143	1.145
	GFLOPs	0.623	0.194	0.176	**0.147**	0.212
	AUC(P_f, P_d)	0.9909	0.9953	0.9932	0.9948	**0.9962**
	Time	**1.735**	2.687	3.391	2.414	2.944
Indian Pines	Params/M	1.149	1.147	1.149	1.152	1.149
	GFLOPs	0.644	**0.193**	0.265	0.236	0.206
	AUC(P_f, P_d)	0.9767	0.9961	0.9823	0.9927	**0.9983**
	Time	**1.917**	3.458	4.117	3.039	3.874

TABLE VI

QUANTITATIVE PERFORMANCE RESULTS OF THE TARGET-AWARE TRANSFORMER NETWORK WITH DIFFERENT DEPTHS ON FIVE DATASETS. OPTIMAL AND SUBOPTIMAL RESULTS ARE INDICATED BY BOLD AND UNDERLINED TEXT, RESPECTIVELY

Dataset		$L=1$	$L=2$	$L=3$	$L=4$	$L=5$
San Diego	Params/M	0.276	0.558	0.840	1.122	1.404
	GFLOPs	0.049	0.097	0.146	0.194	0.242
	AUC(P_f, P_d)	0.9899	0.9923	0.9941	0.9970	**0.9975**
	Time	0.793	1.392	2.162	2.608	3.379
Airport	Params/M	0.287	0.569	0.851	1.133	1.415
	GFLOPs	0.050	0.098	0.147	0.196	0.245
	AUC(P_f, P_d)	0.9851	0.9938	0.9962	**0.9993**	0.9947
	Time	0.744	1.389	1.975	2.641	3.215
Urban-I	Params/M	0.276	0.558	0.840	1.122	1.404
	GFLOPs	0.054	0.108	0.161	0.214	0.268
	AUC(P_f, P_d)	0.9983	0.9985	0.9988	**0.9992**	0.9991
	Time	0.747	1.434	2.050	2.945	3.498
Urban-II	Params/M	0.299	0.581	0.863	1.145	1.427
	GFLOPs	0.054	0.107	0.159	0.212	0.265
	AUC(P_f, P_d)	0.9947	0.9956	0.9960	**0.9962**	0.9956
	Time	0.805	1.446	2.090	2.944	3.239
Indian Pines	Params/M	0.303	0.585	0.867	1.149	1.431
	GFLOPs	0.052	0.104	0.155	0.206	0.258
	AUC(P_f, P_d)	0.9951	0.9967	0.9971	**0.9983**	0.9978
	Time	0.899	1.661	2.738	3.874	5.144

SAM loss, the model's comprehensive performance is robustly improved. This result strongly validates that by constraining the spectral geometry, the SAM loss successfully guides the model to generate more reasonably diverse target samples. These samples better simulate real-world spectral variations, resulting in a trained detector with enhanced generalization capability and more robust performance. To provide a more intuitive assessment of the generation quality and the diversity preserved by the proposed SAM loss, we visualize the spectral curves of the real targets and the generated samples on the Urban-I dataset in Fig. 6. As observed, the generated spectra exhibit realistic fluctuations and geometric variations around the prior spectrum, showing a high degree of consistency with the real target distribution. This visual evidence further confirms that the L_{SAM} constraint effectively guides the model to capture the intrinsic spectral variability, thereby preventing the generation of homogeneous samples.

3) Effectiveness Evaluation of Target-Aware Transformer: The effectiveness of the proposed target-aware transformer architecture was assessed through a performance comparison against several mainstream deep learning backbones, including a multilayer perceptron (MLP) [54], a ViT [21], mamba [55], and a U-Net [56]. A fair comparison was maintained by controlling for model size. Specifically, we modified the number of layers for each architecture to align their parameter counts to a comparable level. The experimental results are presented in Table V. The results confirm that our target-aware transformer attains a superior level of detection performance on all five datasets. This demonstrates that the advantage of our method does not stem from the model's parameter volume but rather from its architectural design. By introducing the prior target information as a guide for cross-attention, our network can more precisely focus on the most discriminative spectral features, thereby achieving superior detection results. We evaluated computational efficiency using giga floating point operations (GFLOPs) and inference time. The results indicate that, although our proposed method is not the fastest, its computational overhead is both reasonable and competitive. Its computational overhead is comparable to that of ViT and U-Net, and significantly better than the mamba model. Overall, our proposed target-aware transformer demonstrates an excellent tradeoff between detection accuracy and computational overhead. It achieves state-of-the-art performance with a moderate computational cost, which fully validates the effectiveness of the target-aware design.

Furthermore, to determine the optimal depth for the proposed target-aware transformer, we conducted a sensitivity analysis by varying the number of transformer blocks L from 1 to 5. The results are detailed in Table VI. As observed, increasing the network depth from $L = 1$ to $L = 4$ yields a consistent and substantial improvement in detection accuracy across all datasets. This positive correlation validates our theoretical premise that a deeper architecture effectively facilitates the extraction of more discriminative, high-level spectral features, which is essential for distinguishing targets from complex backgrounds. However, when the depth is further increased to $L = 5$, the performance gains plateau or even exhibit a slight decline in most scenarios. Consistent with our analysis, this suggests that the model capacity at $L = 4$ is sufficient to capture intrinsic spectral variations, whereas excessive depth begins to introduce optimization difficulties and overfitting risks. From the perspective of efficiency, the computational costs increase linearly with the number of layers. Although $L = 5$ achieves a marginal improvement on the San Diego dataset, it incurs a significantly higher computational cost.

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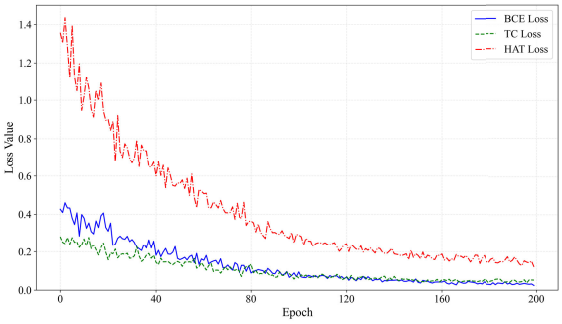


Fig. 7. Convergence curves of the sublosses within the TAM loss during training on the Urban-I dataset.

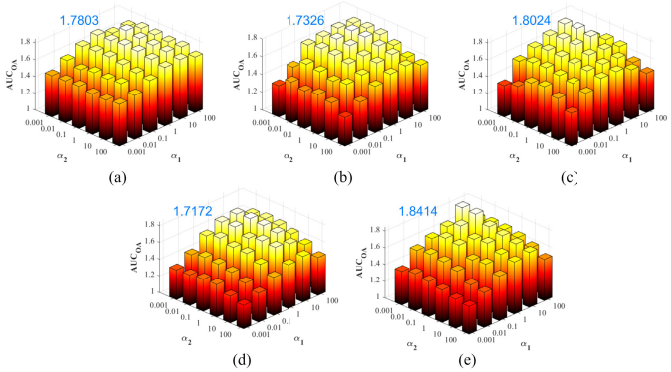


Fig. 8. AUC_{OA} of different datasets across various settings of the parameters α_1 and α_2. (a) San Diego. (b) Airport. (c) Urban-I. (d) Urban-II. (e) Indian pines.

TABLE VII

AUC SCORES OF DIFFERENT LOSS FUNCTION COMBINATIONS USED ON FIVE DATASETS. OPTIMAL AND SUBOPTIMAL RESULTS ARE INDICATED BY BOLD AND UNDERLINED TEXT, RESPECTIVELY

Dataset	L_{HAT}	√	√	√	√
	L_{BCE}		√		√
	L_{TC}			√	√
San Diego	$AUC(P_f, P_d)$	0.9505	0.9832	0.9625	**0.9970**
	$AUC(\tau, P_d)$	0.4902	0.6298	0.6751	**0.7851**
	$AUC(\tau, P_f)$	0.0305	0.0054	0.0183	**0.0018**
Airport	$AUC(P_f, P_d)$	0.9441	0.9812	0.9634	**0.9993**
	$AUC(\tau, P_d)$	0.4163	0.5298	0.6862	**0.7371**
	$AUC(\tau, P_f)$	0.1013	0.0057	0.0092	**0.0038**
Urban-I	$AUC(P_f, P_d)$	0.9087	0.9587	0.9883	**0.9992**
	$AUC(\tau, P_d)$	0.4425	0.4505	0.6405	**0.8042**
	$AUC(\tau, P_f)$	0.0384	**0.0007**	0.0038	0.0010
Urban-II	$AUC(P_f, P_d)$	0.9244	0.9872	0.9881	**0.9962**
	$AUC(\tau, P_d)$	0.5235	0.5493	0.6278	**0.7231**
	$AUC(\tau, P_f)$	0.1445	**0.0015**	0.0073	0.0021
Indian Pines	$AUC(P_f, P_d)$	0.9516	0.9957	0.9677	**0.9983**
	$AUC(\tau, P_d)$	0.4251	0.7184	0.7884	**0.8384**
	$AUC(\tau, P_f)$	0.0754	0.0075	0.0475	**0.0022**

Consequently, setting $L = 4$ strikes the optimal balance between detection accuracy and computational efficiency.

4) Effectiveness Evaluation of Different Loss Terms: To validate the necessity and effectiveness of each component within the TAM loss, we designed four different loss function combinations for an ablation study. As detailed in Table VII, the comparative study reveals the distinct contribution of each loss component to the model. First, the inclusion of either BCE loss or TC loss leads to a substantial improvement in the model's performance. Compared to the baseline model using only the HAT loss, incorporating either the BCE loss or the TC loss alone improves the $AUC(P_f, P_d)$ metric across all datasets. This demonstrates that both the classification supervision provided by the BCE loss and the intraclass compactness constraint from the TC loss are necessary for learning the optimal feature space. Furthermore, the BCE and TC losses have different points of emphasis. The results show that the addition of the BCE loss gives the model an advantage in background suppression, achieving the optimal or suboptimal $AUC(\tau, P_f)$ values on multiple datasets. This highlights the importance of the BCE loss in effectively separating targets from the background. In contrast, incorporating the TC loss leads to superior performance in detection integrity, yielding the suboptimal $AUC(\tau, P_d)$ results across all datasets. This result confirms that the TC loss can effectively counteract spectral variability by encouraging target features to cluster more compactly. Finally, when the complete TAM loss is utilized, the model secures the top $AUC(P_f, P_d)$ values. This indicates that the background suppression capability brought by the BCE loss and the target enhancement effect from the TC loss are complementary, leading to the optimal detection performance. To further verify the effectiveness of each loss component, we visualize their convergence curves on the Urban-I dataset in Fig. 7. The BCE loss declines rapidly in the early stages and achieves stability relatively early, indicating that the pixel-level discrimination capability is effectively learned. The TC loss is consistently minimized throughout the training process, demonstrating that intraclass compactness is explicitly constrained. In contrast, the HAT loss decreases most slowly accompanied by moderate fluctuations, reflecting the model's continuous fine-tuning on hard examples and the gradual enhancement of discrimination ability. These curves validate the rationale of our composite design, where the BCE loss secures pixel accuracy, the TC loss enforces intraclass consistency, and the HAT loss drives the optimization of hard samples.

5) Parameter Analysis: To determine the optimal weighting coefficients α_1 and α_2 in the TAM loss, we employed a grid search strategy. The search space for both parameters was set to the range of 0.001–100. The rationale for selecting this wide range is to accommodate the varying magnitudes of different loss components and to rigorously verify the impact of each term on model performance under varying constraint intensities. In this analysis, we utilize the comprehensive metric AUC_{OA} as the evaluation criterion. Fig. 8 visualizes the grid search results across all five datasets, revealing distinctive performance trends driven by the underlying mechanisms of the loss terms. Regarding the BCE loss weight α_1, the performance generally improves as α_1 increases toward 10. When α_1 is minimal, the explicit classification supervision is insufficient to enforce a sharp decision boundary. However, increasing α_1 to an excessive value of 100 causes the

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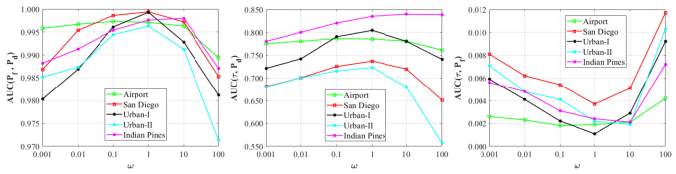


Fig. 9. AUC scores of different datasets across various settings of the parameter ω.

TABLE VIII
GENERALIZATION PERFORMANCE ANALYSIS ON URBAN-I AND URBAN-II DATASETS

Test Data	Train Data	AUC(P_f, P_d)	AUC(τ, P_d)	AUC(τ, P_f)
Urban-I	Urban-I	0.9989	0.8036	0.0012
	Urban-II	0.9873	0.6634	0.0275
Urban-II	Urban-I	0.9835	0.6753	0.0635
	Urban-II	0.9962	0.7231	0.0021

classification task to dominate, potentially forcing the model to prioritize training error minimization at the expense of intrinsic feature learning. In contrast, the TC loss weight α_2 favors a smaller value. Increasing α_2 beyond 1 leads to a sharp performance decline. This deterioration occurs because the model overemphasizes intraclass compactness under such excessive constraints. Consequently, this forces target features to collapse and compromises interclass separability, ultimately degrading overall performance. Consequently, the configuration with $\alpha_1 = 10$ and $\alpha_2 = 0.01$ provides the optimal or near-optimal performance across all datasets. While the Indian Pines dataset attains its highest value of 1.8414 when $\alpha_2 = 0.001$, the performance at $\alpha_2 = 0.01$ remains highly competitive, reaching 1.8345. To ensure robust generalization across diverse scenarios, we select $\alpha_1 = 10$ and $\alpha_2 = 0.01$ as the final configuration. The stable high performance across diverse scenarios, ranging from high-contrast targets in the San Diego dataset to low-contrast subpixel targets in the Indian Pines dataset, demonstrates the robustness of our parameter selection and validates the synergistic design of the TAM loss.

We further investigated the impact of ω in PDM loss, which regulates the balance between L_{cond} and L_{SAM}. Specifically, to examine the impact of varying constraint intensities on overall model performance, we conducted experiments across all five datasets with ω varying from 0.001 to 100, and the results are illustrated in Fig. 9. The results reveal a consistent trend in detection performance across evaluation metrics. When ω is too small, the spectral geometry constraint is insufficient, allowing L_{cond} to dominate the optimization. This causes the generated samples to converge to the prior spectrum, thereby lacking diversity in spectral shape and leading to compromised detection integrity. Conversely, an excessively large ω overemphasizes the geometric constraint, forcing the model to align spectral angles rigidly while neglecting reasonable intensity distributions. This imbalance induces a noticeable degradation in overall performance. Notably, performance peaks consistently around $\omega = 1$ across most datasets, indicating that the PDM achieves an optimal tradeoff between spectral fidelity and shape diversity. Given its robust and stable performance in all experimental scenarios, we set $\omega = 1$ in this study.

6) Generalization Capability Analysis: We evaluated the cross-scenario stability using the Urban-I and Urban-II datasets, which share identical target materials but differ in background composition. To address the spectral dimension mismatch, we aligned the feature spaces by trimming the first and last bands of the Urban-I dataset, utilizing the resulting 205 bands for evaluation. As shown in Table VIII, the model trained on the Urban-II dataset achieves an AUC(P_f, P_d) of 0.9873 on the Urban-I dataset, showing only a marginal decline from the self-trained baseline. Similar robustness is observed in the converse setup, where the model trained on the Urban-I dataset is applied to the Urban-II dataset. Specifically, the high AUC(P_f, P_d) indicates that the model effectively identifies target materials even in a new environment. Although background suppression decreases slightly due to inherent spectral shifts, the consistently high detection integrity confirms that SPDTT captures intrinsic spectral discriminative patterns, demonstrating strong generalization potential.

IV. CONCLUSION AND FUTURE WORK

To mitigate the persistent problems of sample scarcity and spectral variability in HTD, this article proposes a novel HTD approach, termed SPDTT. First, a PDM guided by the prior spectrum and constrained by the SAM loss is designed for data augmentation. Ablation studies demonstrate its superiority in generating high-quality and diverse samples. Second, we develop a target-aware transformer architecture with a cross-attention mechanism. It deeply integrates prior knowledge into the feature extraction process, outperforming several mainstream backbones in extracting discriminative features. Finally, a composite TAM loss is introduced. Through the synergistic optimization of three dedicated components, it learns a feature space with high interclass separability and intraclass compactness. The necessity and effectiveness of each component are validated through ablation studies. Comprehensive experiments conducted on five public datasets indicate that SPDTT achieves state-of-the-art performance by effectively suppressing the background while highlighting targets, and exhibits excellent detection accuracy and generalization capability.

Future work will be pursued in several directions. First, we will explore integrating spatial information for joint spectral–spatial detection. In addition, we aim to incorporate more HSI physical priors to enhance interpretability and reliability. Furthermore, we aim to develop more efficient network architectures for resource-constrained applications.

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